

Project Initialization and Planning Phase

Date	29 June 2026
TeamID	
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution aims to develop the **Voice-Based Concept Understanding Analyser (VBCUA)**, an AI-powered educational assessment system that evaluates users' conceptual understanding through spoken explanations. The platform combines speech-to-text transcription, semantic similarity analysis, audio feature extraction, and intelligent scoring to provide objective feedback on both conceptual knowledge and communication skills. Key features include AI-driven concept evaluation, speech fluency analysis, an interactive Streamlit dashboard, and downloadable PDF performance reports.

Project Overview

Section	Details
Objective	Develop an AI-powered web application that evaluates users' spoken explanations by converting speech into text, measuring semantic similarity with reference concepts, analyzing speech fluency, and generating an overall understanding score with personalized feedback.
Scope	The project enables students, educators, and interview candidates to assess conceptual understanding through voice-based interaction. It includes speech transcription, semantic evaluation, audio feature analysis, waveform visualization, AI-based scoring, and downloadable PDF reports. Future versions may support multilingual analysis, user authentication, and performance tracking.

Problem Statement

Section	Details
Description	Students and learners often struggle to determine whether they have explained a concept correctly. Traditional self-practice lacks objective evaluation, while mentor feedback is not always readily available. Communication issues such as filler words, long pauses, and lack of confidence further reduce the effectiveness of concept explanations during interviews, presentations, and academic assessments.
Impact	Without immediate and personalized feedback, learners may develop misconceptions, lose confidence, and perform poorly in interviews, presentations, and examinations due to weak conceptual understanding and ineffective communication skills.

Section	Details
Approach	Develop an AI-powered application using OpenAI Whisper for speech-to-text transcription, Sentence-BERT for semantic similarity analysis, Librosa for audio feature extraction, and a custom scoring engine to evaluate conceptual understanding and communication quality. The application will be built using Python and Streamlit with a modular architecture.
Key Features	1) AI-powered speech transcription using OpenAI Whisper. 2) Semantic similarity evaluation using Sentence-BERT. 3) Audio analysis including pause ratio, RMS energy, and filler word detection. 4) Intelligent understanding score with qualitative feedback (Strong, Moderate, Poor). 5) Interactive Streamlit dashboard for visualization of transcripts, waveform, and evaluation metrics. 6) Automated PDF report generation using ReportLab. 7) Modular architecture supporting future enhancements such as multilingual support, user authentication, progress tracking, and cloud deployment.

Resource	Purpose
Python	Core application development
Streamlit	Interactive web application
OpenAI Whisper	Speech-to-text transcription
Sentence-Transformers (Sentence-BERT)	Semantic similarity evaluation
Librosa & SoundFile	Audio processing and feature extraction
PyTorch	AI model execution
Matplotlib	Waveform visualization
ReportLab	PDF report generation
NLTK	Text preprocessing
Git & GitHub	Version control and collaboration